

Guiding Generative Probabilistic Weather Models

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Introduction

Motivation

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Extreme events ...

Starting Question

Starting Question

Can we steer generative probabilistic weather models to generate extreme, physically realistic weather events?

① Steer

- Sampling-based guidance
- Guidance-based modification of the generation process

② Extreme

- Defined for a target region and variable(s)
- Evaluated relative to the forecast distribution

③ Physically Realistic

- Realism encouraged by the steering design
- Assessed using realism-related metrics

Goal: Scenario generation for downstream tasks.

Approach

① Steering

- ArchesWeather as the generative probabilistic weather model
- Guidance-based methods adapted from the image domain

② Extreme

- Mask defines the target region and forcing variable
- Change trajectory relative to the unguided forecast distribution
- Focus on heatwave scenarios

③ Physically Realistic

- Guidance is constrained or regularized
- Assessment through:
 - Visual checks
 - Distributional checks using power spectrum

Example

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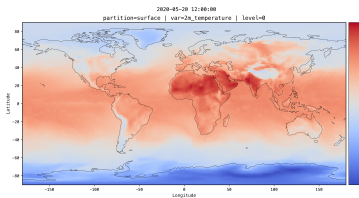
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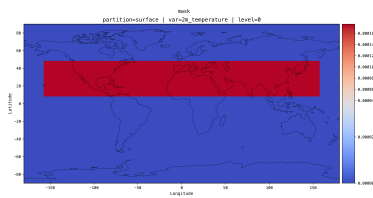
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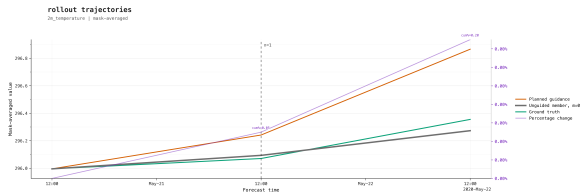
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Target variable



Target region



Guidance signal

Background

Model: ArchesWeather

Forecast decomposition

$$x_{n+1} = \mathbb{E}[x_{n+1} \mid x_n, x_{n-1}, n] + r_{n+1}$$

Implementation

$$\hat{x}_{n+1}^{\text{norm}} = \hat{x}_{n+1}^{\text{det}} + \sigma^{\text{res}} \hat{r}_{n+1}$$

$$\hat{x}_{n+1} = \bar{x}^{\text{data}} + \sigma^{\text{data}} \hat{x}_{n+1}^{\text{norm}}$$

- The deterministic model predicts the conditional mean.
- The flow model generates a residual around this forecast.
- Month, hour, and rollout step are used as time conditioning.

Dataset	ERA5
Forecast step	24h
Grid	global 1.5°, 121 × 240 cells
Variables	4 surface, 6 pressure-level variables
Pressure levels	13 levels, 50–1000 hPa

Flow Matching

- Learned vector field $u_\theta(X_t, t)$ transports noise to data:

$$\frac{d}{dt}X_t = u_\theta(X_t, t)$$

- s.t. sampling generates residuals:

$$r_{n+1} \sim p_{\text{data}}$$

Algorithm 1 Euler Sampling

Require: Vector field u_θ , number of steps T

- 1: Sample $X_0 \sim p_{\text{init}}$
 - 2: $h \leftarrow 1/T$
 - 3: **for** $i = 0, \dots, T - 1$ **do**
 - 4: $t \leftarrow i/T$
 - 5: $X_{t+h} \leftarrow X_t + h u_\theta(X_t, t)$
 - 6: **end for**
 - 7: **return** X_1
-

Weather-guidance design

Guidance Design

- Guidance modifies the sampling process to favor samples with a desired property y .
- In our case, y corresponds to an extreme weather scenario:
 - target variable,
 - target region,
 - desired change over the unguided forecast.
- All guidance methods use this information to modify the vector field:

$$u_t(z_t) \longrightarrow \tilde{u}_t(z_t)$$

- The central design choice is therefore:

Question

How do we define the guidance objective y for weather trajectories?

Weather Guidance Objective

- We guide the generated state toward a prescribed change relative to the unguided forecast.
- The target is defined through a masked spatial average:

Weather Guidance Loss

$$\mathcal{L}_n^{\text{target}} = \left\| \frac{\sum_i m_i \hat{x}_{n,i}}{\sum_i m_i} - (1 + \delta_n) \frac{\sum_i m_i \hat{x}_{n,i}^{\text{ung}}}{\sum_i m_i} \right\|^2$$

- **m**: mask selecting target region and variable
- $\hat{x}_n$: current guided clean-state estimate
- $\hat{x}_n^{\text{ung}}$: unguided prediction
- δ_n : desired relative change at rollout step n

Loss evaluation

The loss is evaluated at different stages during guided sampling depending on the adopted method.

Loss Regularization

- The guidance loss can be regularized to discourage unrealistic deviations from the unguided forecast as follows:

$$\mathcal{L}_n^{\text{total}} = \mathcal{L}_n^{\text{target}} + \lambda \mathcal{R}$$

where $\mathcal{R}$ is the chosen regularization term.

Regularization Choices

$$\mathcal{R}_{\text{ID}} = \|\hat{x}_n - \hat{x}_n^{\text{ung}}\|^2$$

$$\mathcal{R}_{\text{PS}} = \|\text{PS}(\hat{x}_n) - \text{PS}(\hat{x}_n^{\text{ung}})\|^2$$

$$\mathcal{R}_{\text{A}} = \|\text{A}(\hat{x}_n) - \text{A}(\hat{x}_n^{\text{ung}})\|^2$$

- $\mathcal{R}_{\text{ID}}$: keeps the full guided state close to the unguided forecast.
- $\mathcal{R}_{\text{PS}}$: preserves the spatial power spectrum.
- $\mathcal{R}_{\text{A}}$: preserves activity-related structure.

Guiding Weather Rollouts

- Unlike image generation, weather generation evolves over multiple forecast steps.
- The desired change is therefore defined as a trajectory:

$$\delta_1, \delta_2, \dots, \delta_N$$

- At each rollout step n , the guidance objective becomes

$$\mathcal{L}_n = \mathcal{L}(\hat{x}_n, \hat{x}_n^{\text{ung}}, \delta_n, \mathbf{m})$$

- This allows us to prescribe how the extreme event should develop over time.
- We may play with different schemes to define $\{\delta\}_{n=1}^N$.

Notation summary

At rollout step n and flow time t :



- z_t : current flow state
- $\hat{x}_n$: clean estimate at rollout step n at flow time t
- $\mathcal{L}_n^{\text{total}}$: target loss plus optional regularization

Guidance Gradient

$$g_{n,t} = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_n, y)$$

- All methods below use this gradient, but inject it into $\tilde{u}_t^\theta$ using different schemes.

Guidance methods

LBG: Loss-Based Guidance

Idea: modify the vector field using the gradient of a target likelihood.

$$p(x|y) \propto p(y|x)p(x)$$

$$\nabla_x \log p(x|y) = \nabla_x \log p(x) + \nabla_x \log p(y|x)$$

Guided Vector Field

$$\tilde{u}_t(z_t|y) = u_t(z_t) + w a_t \nabla_{z_t} \log p(y|z_t)$$

Using the weather loss:

$$\nabla_{z_t} \log p(y|z_t) = -\nabla_{z_t} \mathcal{L}_n(\hat{x}_n, y) = g_{n,t}$$

- For linear probability paths: $a_t = \frac{1-t}{t}$
- w controls guidance strength
- For $w \neq 1$, the exact probabilistic interpretation is lost
- Unlike DPS paper, we do not rescale $w a_t$ by $\|\nabla_{z_t}\|$.

LBG-MC: Loss-Based Guidance Monte Carlo

Idea: avoid guiding only through the point estimate $\hat{x}_n := x(\hat{z}_0)$.

- Account for uncertainty in the clean state:

$$\begin{aligned} p(y|z_t) &= \int p(y|\hat{x}_n)p(\hat{x}_n|z_t) d\hat{x}_n \\ &= \mathbb{E}_{\hat{x}_n|z_t} [\exp(-\mathcal{L}_n(\hat{x}_n, y))] \end{aligned}$$

- Since the true posterior $p(\hat{x}_n|z_t)$ is intractable, approximate it by

$$q(\hat{x}_n|z_t) = \mathcal{N}(\hat{x}_n, \sigma_t^2 I)$$

- In our implementation:

$$\sigma_t = R a_t$$

and we perturb $\hat{x}_n^{\text{norm}}$ instead of $\hat{x}_n$ directly.

LBG-MC: Loss-Based Guidance Monte Carlo

Sample clean-state candidates:

$$x^{(i)} = \hat{x}_n + \sigma_t \varepsilon^{(i)}, \quad \varepsilon^{(i)} \sim \mathcal{N}(0, I), \quad \sigma_t = R a_t$$

Approximate the guidance likelihood:

$$p(y|z_t) \approx \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} \exp\left(-\mathcal{L}_n(x^{(i)}, y)\right)$$

$$\nabla_{z_t} \log p(y|z_t) \approx \nabla_{z_t} \log \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} \exp\left(-\mathcal{L}_n(x^{(i)}, y)\right)$$

Monte-Carlo Loss

$$\tilde{\mathcal{L}}_n = \log N_{\text{MC}} - \log \text{sumexp}_i \left(-\mathcal{L}_n(x^{(i)}, y) \right)$$

- N_{MC} : number of Monte-Carlo samples
- R : spread magnitude; a_t shrinks the cloud over the flow
- Limitation: Gaussian posterior approximation

UG: Universal Guidance

Idea: optimize noisy state before taking the generative step.

Inner Optimization

$$\Delta z_t^* = \arg \min_{\Delta z} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z), y)$$

In practice, approximate Δz_t^* with K gradient steps:

$$\Delta z_t \leftarrow \Delta z_t - \eta \nabla_{\Delta z} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z), y)$$

Then apply the optimized correction:

$$z_t^{\text{gui}} = z_t + w a_t \Delta z_t$$

Strong correction may leave the data manifold $\rightarrow$ re-inject noise.

Bansal et al., Universal Guidance for Diffusion Models, CVPR Workshop 2023.

UG: Universal Guidance

Idea: repeat guidance at the same flow time t before moving to $t + h$, in a self recurrent fashion.

One Self-Recurrence Step

$$\begin{aligned} z_t^{(s)} &\xrightarrow{\text{guide}} z_t^{(s),\text{gui}} = z_t^{(s)} + w a_t \Delta z_t^{(s)} \\ z_t^{(s),\text{gui}} &\xrightarrow{\text{Euler}} z_{t+h}^{(s)} = z_t^{(s),\text{gui}} + h u_t(z_t^{(s),\text{gui}}) \\ z_{t+h}^{(s)} &\xrightarrow{\text{renoise}} z_t^{(s+1)} \end{aligned}$$

- Repeat this loop for $s = 1, \dots, S$ at the same flow time t .
- Re-noising brings the sample back to the same flow level after a "trial" step.
- After the last recurrence, keep the Euler step and move to $t + h$.
- Hyperparameters: K inner optimization steps, S recurrence steps, guidance strength w , learning rate η

Universal Guidance: Algorithm

Algorithm 2 Universal Guidance with Self-Recurrence

Require: Vector field u_θ , loss $\mathcal{L}_n$, clean predictor $\hat{x}_{n,t}$, steps T , inner steps K , recurrences S

```
1: Sample  $z_0 \sim p_{\text{init}}$ 
2:  $h \leftarrow 1/T$ 
3: for  $j = 0, \dots, T - 1$  do
4:    $t \leftarrow j/T$ 
5:   for  $s = 1, \dots, S$  do
6:      $\Delta z_t \leftarrow 0$ 
7:     for  $k = 1, \dots, K$  do
8:        $\Delta z_t \leftarrow \Delta z_t - \eta \nabla_{\Delta z_t} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z_t), y)$ 
9:     end for
10:     $z_t^{\text{gui}} \leftarrow z_t + w a_t \Delta z_t$ 
11:     $z_{t+h} \leftarrow z_t^{\text{gui}} + h u_\theta(z_t^{\text{gui}}, t)$ 
12:    if  $s < S$  then
13:       $z_t \leftarrow \text{renoise}(z_{t+h}, t + h \rightarrow t)$ 
14:    end if
15:  end for
```

FlowGrad (FG)

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FlowGrad-free (FGF)

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Runtime Overview

Method	Mechanism	Cost / flow step
LBG	loss gradient at $\hat{x}_n$	1 backward
LBG-MC	MC-averaged likelihood gradient	S backward
UG	inner-loop latent shift Δz_t	$K \times S$ backward
FG	optimized schedule λ_t on $g_{n,t}$	backprop through ODE
FGF	free controls c_t	backprop through ODE

- **Score-based:** LBG and LBG-MC add a guidance gradient to the vector field.
- **Optimization-based:** UG, FG, and FGF optimize correction variables during sampling.

Method-Specific Hyperparameters

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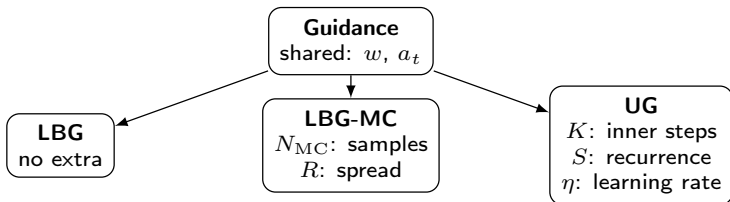
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Experimental Setup

Parameter	Meaning
Start time	input-state timestamp at $n = 0$
M	number of ensemble members
N	number of rollout steps
T	number of flow steps per rollout step
Reference	unguided forecast
$\{\delta_n\}_{n=1}^N$	prescribed target change trajectory
λ	regularization strength
$\mathcal{R}$	regularization type
Mask	target region and variable
Mask mode	BBOX or Gaussian mask

Evaluation framework

For each guidance method, we report following:

- **Best hyperparameter setting:** the setting that best balances target achievement and physical realism.
- **Target achievement:** how closely the guided rollout follows the prescribed δ_n trajectory.
- **Realism cost:** how much the rollout deviates from the unguided forecast distribution.
- **Stability:** whether the method produces smooth and stable rollouts across forecast steps.
- **Runtime cost:** the additional computational cost compared to unguided sampling.
- **Findings:** trade-off between target achievement and realism, or optimal guidance schedule.

Evaluation Metrics

We evaluate the guidance methods using two main questions:

- Can the method achieve the prescribed guidance target?
- Are the resulting guided rollouts physically meaningful?

We quantify these questions with two main metrics:

- **Target error:** distance between the realized guided change and the prescribed target trajectory.
- **Spectral deviation:** distance between the guided- and the unguided power spectrum.

In addition, we inspect the guided rollouts qualitatively through:

- final weather maps,
- guidance gradients,
- modified vector fields,
- ...

Power Spectrum: Spherical-Harmonic Expansion

We compute the **spherical harmonics** for each variable slice $x_n^v(\theta, \varphi)$

Spherical-harmonic expansion

$$x_n^v(\theta, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} a_{\ell m}^v Y_{\ell m}(\theta, \varphi)$$

- x_n^v : a single (θ, φ) field — a surface variable or one pressure level.
- $Y_{\ell m}$: spherical harmonic of degree ℓ , order m — the orthonormal basis on the sphere.
- $a_{\ell m}^v$: expansion coefficient, the projection of x_n^v onto $Y_{\ell m}$.
- Each degree ℓ carries $2\ell + 1$ orders $m = -\ell, \dots, \ell$ (the orientations of that scale).
- $L = 59$: max degree for our 120×240 grid ($\ell = 0, \dots, 59$).

Spectral Realism Metrics

Per slice x_n^v , we compare the guided spectrum $\text{PS}_n^{\text{gui}}(\ell)$ to a reference $\text{PS}_n^{\text{ref}}(\ell)$ (unguided forecast or ground truth), over the L degrees $\ell \geq 1$ ($\ell = 0$, the field mean, is dropped).

Log-spectral distance — magnitude of mismatch

$$\text{LSD}_n = \sqrt{\frac{1}{L} \sum_{\ell \geq 1} \left(\ln \text{PS}_n^{\text{gui}}(\ell) - \ln \text{PS}_n^{\text{ref}}(\ell) \right)^2}$$

Spectral bias — signed direction

$$\text{bias}_n = \frac{1}{L} \sum_{\ell \geq 1} \ln \frac{\text{PS}_n^{\text{gui}}(\ell)}{\text{PS}_n^{\text{ref}}(\ell)} \quad \begin{cases} < 0 & \text{too smooth} \\ > 0 & \text{too rough} \end{cases}$$

Total power — structure magnitude

$$P_n = \sum_{\ell \geq 1} \text{PS}_n(\ell)$$

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QA

Ideas

- Explain why not just do M-step GD on the deterministic prediction.

Appendix

Background

Flow Matching

- Assume observations are samples from a data distribution

$$p_{\text{data}}$$

- Generation corresponds to sampling from this distribution
- Start from a simple distribution

$$p_{\text{init}}$$

- Learn a transformation

$$p_{\text{init}} \longrightarrow p_{\text{data}}$$

Representing the Transformation

- The transformation is represented by a **vector field**

$$(t, x) \mapsto u_t(x)$$

- The vector field defines a trajectory

$$t \mapsto x_t$$

- Through the ODE

$$\frac{d}{dt}x_t = u_t(x_t)$$

- Solving the ODE induces a flow

$$x_t = \phi_t(x_0)$$

Sampling

- Learn a neural vector field

$$u_{\theta}(x_t, t)$$

- Solving

$$\frac{d}{dt}x_t = u_{\theta}(x_t, t)$$

transports samples from

$$p_{\text{init}} \rightarrow p_{\text{data}}$$

- The solution is generally not available in closed form.
- Numerical integration is therefore required.

Euler Integration

Algorithm 3 Sampling from a Flow Model

Require: Vector field u_θ , number of steps T

- 1: Sample $X_0 \sim p_{\text{init}}$
 - 2: $h \leftarrow 1/T$
 - 3: **for** $i = 0, \dots, T - 1$ **do**
 - 4: $t \leftarrow i/T$
 - 5: $X_{t+h} \leftarrow X_t + h u_\theta(X_t, t)$
 - 6: **end for**
 - 7: **return** X_1
-

From Sampling to Training

- During sampling, the learned ODE transports

$$X_0 \sim p_{\text{init}} \longrightarrow X_1 \sim p_{\text{data}}$$

- To train the vector field, we prescribe how intermediate samples should look:

$$p_0 = p_{\text{init}} \longrightarrow p_t \longrightarrow p_1 = p_{\text{data}}$$

- This family of intermediate distributions is the **probability path**.
- Flow matching trains u_θ to match the velocity of this path.

Training Flow Matching

- Define a Gaussian probability path conditioned on a sample x_1 :

$$p_t(x \mid x_1) = \mathcal{N}(x; tx_1, (1-t)^2 I)$$

- Equivalently, sample along this path using

$$x_t = (1-t)x_0 + tx_1, \quad x_0 \sim \mathcal{N}(0, I), \quad x_1 \sim p_{\text{data}}$$

- The target velocity along this path is

$$u_t^*(x_t) = \frac{d}{dt}x_t = x_1 - x_0$$

- Train u_θ to match this velocity:

$$\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2 \right]$$

From Loss to Guidance Gradient

- Once the target and regularization terms are chosen, the total loss defines the guidance signal:

$$\mathcal{L}_n^{\text{total}} = \mathcal{L}_n^{\text{target}} + \lambda \mathcal{R}$$

- We define the guidance likelihood through this loss:

$$p(y|z_t) = \exp(-\mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y))$$

- Therefore, the guidance gradient is:

$$\nabla_{z_t} \log p(y|z_t) = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y)$$

Guidance Signal

$$g_{n,t} = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y)$$

LBG: Loss-Based Guidance

Idea: same loss-gradient guidance as classifier guidance, but with a normalized guidance scale.

LBG Update

$$\tilde{u}_t(z_t) = u_t(z_t) + \psi_t g_{n,t}, \quad g_{n,t} = -\nabla_{z_t} \mathcal{L}_n(\hat{x}_{n,t}, y)$$
$$\psi_t = \frac{w a_t}{\|g_{n,t}\| + \varepsilon}$$

- Hyperparameters: guidance strength w .
- $\hat{x}_{n,t}$ is treated as a point estimate of the clean state.
- Normalization stabilizes guidance magnitude across flow steps.
- Limitation: ignores uncertainty in $x_1|z_t$.

Chung et al., Diffusion Posterior Sampling for General Noisy Inverse Problems, ICLR 2023.